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Alzheimer's Disease Research



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GraphSAGE

Mean aggregation · NeurIPS 2017

Connectivity profile of ROI i

$$x_i = [\text{corr}(i, R_1), \dots, \text{corr}(i, R_N)]$$

Mean of neighboring node features

$$m_i = \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} x_j$$

Updated node embedding

$$h_i = \sigma(W_{\text{self}}x_i + W_{\text{neigh}}m_i + b)$$

Notation

$\mathcal{N}(i)$: neighbors of ROI i

$W_{\text{self}}, W_{\text{neigh}}$: learned weights

b : learned bias

σ : activation function

Layer output

$$H = \begin{pmatrix} h_1^T \\ \vdots \\ h_N^T \end{pmatrix} \in \mathbb{R}^{N \times d}$$

N nodes, d embedding features

The Model

Functional connectivity graphs across a subject's visits

